

Algorithmic Pseudocode Sample 23

Source-backed Image2Struct algorithm sample

1 Algorithm

This sample contains algorithmic pseudocode extracted from a source-backed LaTeX benchmark dataset. It is wrapped in a minimal article document for pdfLaTeX validation.

Algorithm 1 Source-backed algorithmic procedure

Require: Proteins $\mathcal{P} = \{P_i\}_{i=1}^N$, PPIs set $\mathcal{X}$, pre-training epoch E_{pre} , and training epoch E .

- 1: Randomly initializing the parameters of protein encoder f_e , decoder f_d , and codebook $\mathcal{A}$.
- 2: *# Pre-training*
- 3: **for** epoch $\in \{1, 2, \dots, E_{pre}\}$ **do**
- 4: Encoding residue embeddings $\{\mathbf{h}_m^{(L)}\}_{m=1}^M$ for each protein by the encoder f_e in Eq. (??);
- 5: Performing vector quantization by the codebook $\mathcal{A}$ in Eq. (??);
- 6: Reconstructing the input from discrete codes $\{z_m\}_{m=1}^M$ by the decoder f_d ;
- 7: Masking the codebook by Eq. (??) and reconstructing the input from the masked $\{\tilde{\mathbf{e}}_{z_m}\}_{m=1}^M$;
- 8: Optimizing encoder f_e , decoder f_d , and codebook $\mathcal{A}$ by minimizing the loss in Eq. (??).
- 9: **end for**
- 10: *# PPI Prediction*
- 11: Freezing the parameters of encoder f_e and codebook $\mathcal{A}$.
- 12: Randomly initializing the parameters of PPI encoder g_{enc} and PPI classifier g_{cla} .
- 13: **for** epoch $\in \{1, 2, \dots, E\}$ **do**
- 14: Encoding all proteins into microenvironment-aware protein embeddings by Eq. (??);
- 15: Updating node (protein) embeddings on the PPI graph by the PPI encoder g_{enc} in Eq. (??);
- 16: Predicting unknown PPI types by the PPI classifier g_{cla} .
- 17: Optimizing PPI encoder g_{enc} and PPI classifier g_{cla} by minimizing the objective in Eq. (??).
- 18: **end for**
- 19: **return** Predicted PPI types $\mathcal{Y}_U$, protein encoder f_e , and microenvironment codebook $\mathcal{A}$.
